Mathematical Modeling of Tridosha Homeostasis Using Dynamical Systems and Feedback Control Theory

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This project formalizes Vata, Pitta and Kapha as a three-component composition evolving around a fixed equilibrium. A continuous dynamical system describes deviations from that equilibrium, a bounded negative-feedback controller opposes those deviations, and numerical experiments quantify stability, perturbation response and recovery. The deliverable includes a Python model, a local interactive website, reproducible experiments, fitted cohort-classifier coefficients, regression tests and the preserved input codebase.

The central result is mathematical: the proposed unforced model has a globally asymptotically stable equilibrium under its stated parameter restrictions. In the default illustrative experiment, feedback reduces sampled post-pulse recovery time from 6.4 to 2.6 days and reduces integrated deviation by approximately 49.6%. These are simulator results, not patient outcomes. The time unit is an illustrative day; no longitudinal clinical data were supplied to estimate real recovery rates.

The ML work is deliberately separate. A synthetic parameter-recovery experiment verifies that the calibration procedure can recover known decay rates under controlled conditions. A cohort classifier benchmarks extreme constitution labels using the supplied cross-sectional data. Neither experiment demonstrates treatment effectiveness or clinical validation of the ODE model.

1 Project scope and delivered features

The project title determines the active scope: coupled ODEs, homeostatic equilibrium, perturbations, negative feedback, stability and recovery. The supplied planning document describes a broader tracking application. It is treated as design context rather than an instruction to implement every suggested feature. Its fixed Prakriti reference and compositional geometry are retained as mathematical concepts.

The active website provides four starting scenarios: Vata, Pitta or Kapha perturbation, and equilibrium. Users set equilibrium weights, pulse strength, pulse start and duration, simulation horizon and feedback gain. Advanced controls expose the three natural decay rates and coupling strength. Each experiment compares the same initial condition and forcing with feedback enabled and disabled.

Outputs include the two local stability eigenvalues, a recovery-time estimate, cumulative imbalance reduction, a full distance-versus-time chart, a ternary trajectory and a selectable time snapshot. The snapshot shows all three proportions and their Aitchison distance from equilibrium. Export saves the full completed experiment as JSON, including the configuration used to calculate it. A separate evidence section displays actual numerical verification, synthetic calibration and cohort benchmark results generated by Python.

The original project remains under baseline_v2, including the questionnaire, daily logs, environmental mappings, weather adapter, herb lookup and historical version. These are retained for traceability, not connected as additional features of the new research website. No account system, clinical diagnosis, treatment recommender, food analysis, language model or new health-logging module has been added.

The deliverable is local and self-contained at the source level. It needs Python dependencies installed once, but the active simulator makes no external weather calls and needs no API key. No user profiles or personal health information are stored by the new service. This boundary keeps the implementation aligned with a mathematical modelling project rather than a medical product.

2 Review of the supplied baseline

The supplied v2 model represents composition as closure(p * exp(x)), with fixed reference p and a centered deviation x. Its daily recurrence is x_next = center(c * tanh((diag(rho) * x + f) / c)). The recurrence includes lifestyle, environmental and digestive terms. It is a discrete-time model, not a continuous ODE. The previous structure is useful for daily tracking, but it does not by itself supply the continuous-time stability and feedback-control analysis named in the title.

The original code includes transparent Dirichlet questionnaire scoring, draft expert mapping constants, a softmax constitution classifier and cross-cohort validation. Several assumptions are explicitly marked as requiring practitioner review. Their numerical values should not be presented as established biological constants.

The new implementation makes the coupling explicit, adds an explicit state-dependent controller, separates forced and unforced dynamics, and analyzes the physically relevant two-dimensional tangent space. It preserves the desirable fixed-equilibrium and compositional properties without treating a daily recurrence as an ODE.

The original cohort preprocessing constructs its category vocabulary using both cohorts. This is target-covariate access, even though target labels are not used for the vocabulary. The revised benchmark learns categories entirely inside the training pipeline and handles unseen target categories by all-zero encoding. Hyperparameter selection is also confined to the training cohort.

Other baseline limitations remain visible in the preserved source: state() indexes an empty trajectory before any log exists; previous-day output can mean the previous logged observation rather than the previous calendar day; history-dependent energy summaries can depend on edit order; and weather tests may use live services or a region-specific fallback. These paths are not imported by the new server. The archive is preserved rather than silently rewritten or certified as a finished tracking product.

Eleven baseline tests that do not require the live API/forecast paths were rerun successfully. Two original tests were deselected because they exercise weather-dependent behavior. The active v3 model and API have their own deterministic tests, discussed in Section 12. Historical legacy_v1 tests were not part of the new application acceptance suite.

3 State variables and compositional geometry

Let p = (p_V, p_P, p_K) be the equilibrium composition, with p_i > 0 and sum(p_i) = 1. These are normalized model coordinates. They are not measured concentrations or percentages of a known biochemical substance. The website accepts positive baseline weights and normalizes them once per experiment.

Let x belong to the plane H = {x in R^3 : sum(x_i) = 0}. Each component is a centered log-ratio deviation from equilibrium. The time-dependent composition is defined by the baseline-weighted softmax:

```
v_i(t) = p_i exp(x_i(t)) / sum_j[p_j exp(x_j(t))]
P = I - (1/3) 1 1^T
x(0) = P x_input
```

The implementation uses scipy.special.softmax(log(p) + x), which subtracts the largest exponent internally for numerical stability. Every finite state maps to strictly positive shares summing to one. At x = 0, v = p. A common offset added to all initial components has no effect because the projection removes it.

For a positive composition z, clr(z) = log(z) - mean(log(z)). Substitution gives clr(v) - clr(p) = x on H. Consequently the Aitchison distance is exactly d_A(v,p) = ||x||_2. This provides a consistent imbalance metric without applying Euclidean distance directly to closed proportions. A value such as 0.05 is a numerical threshold chosen for the experiment, not a clinically validated boundary.

The ternary diagram places Vata at the lower left, Pitta at the lower right and Kapha at the top. Cartesian coordinates are X = v_P + v_K/2 and Y = sqrt(3) v_K/2. The display rescales these coordinates for an SVG drawing. The equilibrium diamond, initial hollow circle and selected filled circle distinguish three different states; the line shows the controlled trajectory through composition space.

There are only two independent coordinates despite three named doshas. All stability calculations must respect this constraint. Reporting a zero eigenvalue associated with a redundant direction as a physiological mode would be misleading. The implementation therefore works with an orthonormal basis Q for H when computing eigenvalues.

4 Coupled ODE and negative feedback

Define D = diag(r_V, r_P, r_K), with each r_i positive. Natural decay represents passive relaxation of deviations. Define L as the complete-graph Laplacian with diagonal entries 2 and off-diagonal entries -1. The coupling coefficient c is nonnegative. The continuous model is:

```
dx/dt = P [-D x - c L x + f(t) - k s tanh(x/s)]
L = [[ 2, -1, -1], [-1, 2, -1], [-1, -1, 2]]
f(t) = a e_j   for t_start <= t < t_start + duration
f(t) = 0       otherwise
```

The nonlinear function is applied componentwise. The raw control signal is u_raw = -k s tanh(x/s), where k >= 0 and s > 0. It is opposing feedback: a positive deviation generates a negative correction and a negative deviation generates a positive correction. For small deviations u_raw is approximately -k x; for large deviations each raw component approaches a finite magnitude k s.

The actual centered input is P u_raw. Projection can increase an individual component relative to the raw bound, so the safe componentwise bound is 2 k s for P u_raw, not k s. The implemented control-effort metric uses this projected input. A regression test checks both dissipation and the projected bound.

Default parameters are r = (0.30, 0.22, 0.14) day^-1, c = 0.06 day^-1, k = 0.45 day^-1 and s = 0.8 log-ratio units. The default baseline is (0.34, 0.33, 0.33), the initial deviation is (0.85, -0.30, -0.55), and a Vata-directed pulse of strength 0.6 acts from day 3 to day 5. The simulation ends at day 21.

Coupling is deliberately symmetric and phenomenological. It formalizes interaction but does not identify a biological mechanism. In this complete-graph model Lx = 3x on H, so coupling acts as isotropic additional damping in the tangent plane. A more elaborate network would introduce extra unvalidated parameters without being needed for the requested core idea.

The equivalent composition ODE follows by differentiating the softmax: dv_i/dt = v_i [dx_i/dt - sum_j(v_j dx_j/dt)]. It sums to zero and preserves the simplex for finite solutions. The implementation integrates x rather than v because positivity is then enforced by construction, avoiding ad hoc clipping of negative proportions.

5 Equilibrium and stability derivation

With f = 0, x = 0 is an equilibrium and corresponds to v = p. Use V(x) = 0.5 x^T x as a Lyapunov function on H. Since Px = x and P is symmetric, projection disappears when taking the inner product with x:

```
dV/dt = -x^T D x - c x^T L x
        - k s sum_i [x_i tanh(x_i/s)]
```

The first term is strictly negative for any nonzero x because every decay rate is positive. The second is nonpositive because L is positive semidefinite. Each product x_i tanh(x_i/s) is nonnegative, so the controller cannot inject Lyapunov energy in the unforced model. Thus dV/dt < 0 away from the equilibrium. The vector field is globally Lipschitz and V is radially unbounded on H; the equilibrium is globally asymptotically stable.

In fact, x^T Lx = 3 ||x||^2 on H. Hence dV/dt <= -2 (min(r) + 3c) V. This yields the conservative bound ||x(t)|| <= ||x(0)|| exp[-(min(r)+3c)t] after forcing ends. For the defaults, the bound uses 0.32 day^-1. Feedback improves local decay, but its saturation means the global proof should not simply add k to this uniform linear bound.

An orthonormal basis is Q = [(1,-1,0)/sqrt(2), (1,1,-2)/sqrt(6)]. The local tangent Jacobian is A_H = -Q^T(D + cL + kI)Q. Its eigenvalues are real and negative because the matrix is symmetric negative definite. Default feedback eigenvalues are approximately -0.896188 and -0.803812 day^-1. Without feedback they are -0.446188 and -0.353812 day^-1.

The slow local time constants are about 1.244 and 2.826 days respectively, computed as -1/lambda_max. These are local linearized time constants, not the same as a finite-threshold settling time after a pulse. During a persistent disturbance, the reference p is generally not an equilibrium. The field remains dissipative and a forced equilibrium is shifted; returning to p is assessed only after the pulse ends.

For bounded forcing, x^T Pf <= ||x|| ||Pf||. This implies an ultimate norm bound based on forcing magnitude divided by min(r)+3c. It explains why finite forcing produces a bounded response, but does not establish that all possible control gains optimize every performance measure or correspond to safe real interventions.

6 Numerical simulation and measurement

The solver is scipy.integrate.solve_ivp using its adaptive RK45 method. Relative tolerance is 1e-9, absolute tolerance is 1e-11, and maximum internal step is 0.25 day. The returned reporting grid normally has 0.1-day spacing, augmented with the exact start and end of the pulse and the final time. Adaptive internal steps and reporting samples are different quantities.

The integration is split at each forcing discontinuity. Within a segment the pulse is constant, so the solver never steps blindly across a jump in forcing. The final state of one segment becomes the initial state of the next. Dense output evaluates each segment on the common reporting grid. A final projection removes accumulated floating-point centering error.

The numerical verification experiment disables feedback and forcing. In the tangent plane the exact linear solution is x(t) = Q exp(A_H t) Q^T x(0). Comparing this matrix-exponential solution with the integrated trajectory gives a maximum absolute component error of approximately 1.30e-10 in the default experiment. This verifies a tractable limit of the solver/model implementation; it is not a direct error guarantee for every nonlinear setting.

Recovery time is the earliest sampled time after the pulse ends at which distance is at most 0.05 and remains so at all later samples through the selected horizon. The reported value is elapsed time since pulse termination. A null value means the criterion was not reached within the horizon, and the website displays that explicitly. The measurement is quantized to the reporting grid and does not establish behavior beyond the finite observation window.

Integrated imbalance is the trapezoidal integral of ||x(t)|| over the full simulation. Control effort is the trapezoidal integral of ||P u_raw(t)||^2. Peak distance includes the initial condition. The reported percentage reduction compares controlled and natural integrated imbalance under identical initial conditions and forcing. If the natural integral is essentially zero, the UI displays a dash rather than dividing by zero.

The API rejects nonfinite values, invalid baseline dimensions or sums, nonpositive decay, negative gain/coupling, invalid pulse windows and excessive horizons. The website bounds input controls too, but server validation remains authoritative. The public API has a 90-day horizon limit and no option to request arbitrarily fine output resolution.

7 Numerical results and controller tradeoffs

The default comparison yields the following measured quantities. Both systems start with distance 1.055936. The controlled trajectory ends at approximately 4.53e-7, while natural recovery ends at 0.00106446. Post-pulse recovery time is 2.6 days with feedback and 6.4 days without feedback. Integrated imbalance is 2.373053 versus 4.710985, a reduction of approximately 49.6%. Controlled effort is 0.177502, compared with zero when the controller is disabled.

The gain sweep keeps all other parameters fixed:

Feedback gain	Recovery days	Integrated imbalance	Control effort
0.00	6.4	4.710985	0.000000
0.15	4.4	3.552263	0.032920
0.45	2.6	2.373053	0.177502
0.90	1.5	1.582381	0.432123
1.50	0.9	1.095492	0.775847

In this experiment stronger feedback produces faster recovery and lower integrated deviation, at the cost of more control effort. A model cannot choose an optimal treatment simply from this curve: there is no measured treatment-cost function, toxicity model, actuator delay or clinical outcome data. Gain is an experimental control parameter, not a dosage.

A nine-case sensitivity grid varies all decay rates together by factors 0.8, 1.0 and 1.2 and coupling across 0.03, 0.06 and 0.09. Each case compares the controlled system with its own natural baseline. The complete measured results are stored in results/experiments.json. Across this grid, controlled recovery spans 2.1-3.3 days and integrated-imbalance reduction spans 42.7-59.2%. These are scenario sensitivity results, not probability-based confidence intervals: the grid is not a distribution fitted to observations.

The model's guaranteed stability follows from its sign restrictions, not from selecting only successful simulations. Because every case maintains positive decay, nonnegative coupling and nonnegative gain, the unforced proof continues to apply. Timing and magnitude still depend on parameter choices; stability alone does not establish physiological realism.

The recovery and composition figures included with the report are generated from results/default_trajectory.json. They show the same Python outputs that the website renders. The default configuration in the browser was aligned with the experiment script so that displayed summary values and report values correspond.

Numerical recovery trajectories

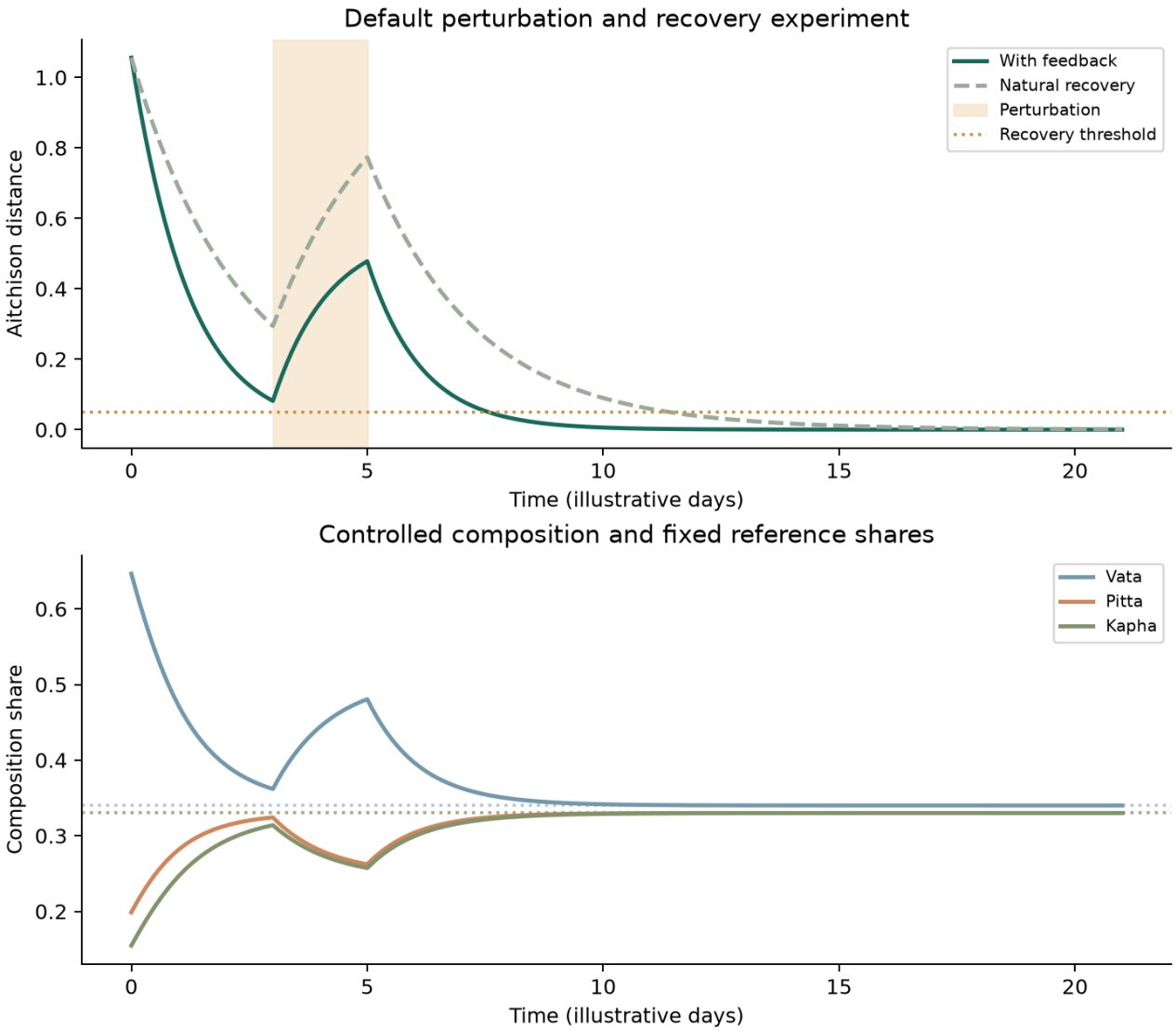


Figure 1. Default Python simulation. Shaded interval: pulse from day 3 to day 5. Dotted horizontal composition lines indicate the fixed equilibrium. The controller changes the path and recovery rate; it does not change Prakriti.

8 System identification and synthetic calibration

The supplied data do not include repeated dosha-state observations and known perturbations over time. It would therefore be unjustified to claim that the ODE decay rates were learned from real recovery records. Instead, a synthetic parameter-recovery experiment checks whether the proposed fitting procedure behaves correctly when the generating system is known.

Three training trajectories use different centered initial states and pulse targets. Each lasts 10 days, with a pulse of strength 0.5 from day 2 to day 4 and observations every 0.5 day. The generating decay rates are (0.30, 0.22, 0.14), coupling is fixed at 0.06, and feedback is disabled. Independent Gaussian component noise of standard deviation 0.004 is projected onto H before being added to the state observations. The random seed is 91.

Bound-constrained nonlinear least squares minimizes the stacked residuals between simulated and noisy observed states. The initial guess is (0.5, 0.5, 0.5), and bounds for each fitted decay rate are 0.03 to 1.0. The fitted rates are approximately (0.300285, 0.220683, 0.139166). The optimizer reports success and a local residual-Jacobian rank of three.

A held-out deterministic trajectory uses initial state (0.6, 0.2, -0.8), a Pitta pulse of strength 0.8 from day 4 to day 7, and a 14-day horizon. State RMSE falls from 0.137591 under the initial guess to 0.000304772 after fitting. This is a controlled synthetic recovery result, not evidence of generalization to humans or proof that noisy clinical observations would identify the parameters equally well.

Identifiability is important here. On H, the complete-graph coupling contributes 3cI. Simultaneously adding an equal amount to all decay rates and subtracting the corresponding amount from c leaves the linear field unchanged. Thus fitting all three decay rates and coupling together would be structurally confounded. The experiment fixes coupling and fits only the decay rates. Feedback gain and saturation are also held fixed rather than attempting to estimate every parameter from the same limited data.

For future empirical calibration, observations would need a defensible measurement model, repeated assessments, known or recorded inputs, missingness handling and subject-level train/test separation. A state-space or observation-noise formulation could then be justified. The current code does not implement the planning document's proposed Kalman-filter personalization or weekly check-in feature.

9 Cohort data and ML methodology

The benchmark uses the unmodified Western_India.csv and Northern_India.csv supplied in the original archive. They contain 131 and 86 records respectively, giving 217 records in total. Schema alignment identifies 105 shared feature columns after excluding SampleID and Class. The class labels are Vata, Pitta and Kapha, representing the extreme constitution groups in this dataset. Mixed constitution shares and longitudinal Vikriti are not targets.

The original attribution names the Mendeley dataset with DOI 10.17632/dyx5cgxdrf.1 and CC BY 4.0, linked to Tiwari et al. (2017). The packaged attribution files are retained. The feature dictionary in the input archive is marked inferred; it is not promoted into a validated questionnaire. The study's broader population counts should not be substituted for the actual 217 rows used here.

For each direction, one entire geographic cohort is used for development and the other only for final evaluation. Missing entries are replaced with the literal category __missing__. A scikit-learn Pipeline contains OneHotEncoder(handle_unknown='ignore') and regularized multinomial logistic regression. All categories are learned inside each training fold. No target-cohort frequencies, feature-value categories or labels are used to fit preprocessing or choose regularization.

The candidate inverse regularization strengths are C = 0.01, 0.1, 1 and 10. Four-fold stratified cross-validation uses shuffle=True and random_state=42 on the training cohort only. Selection optimizes negative log loss rather than accuracy. The selected pipeline is refitted on the complete training cohort and evaluated once on the held-out cohort. The small candidate grid limits unnecessary selection flexibility; it does not eliminate all small-sample uncertainty.

Comparator models are a training-majority classifier and a fixed-C logistic model with C = 0.1 using the same training-only encoding. The fixed-C comparison is explicitly not a bit-for-bit replay of the original custom softmax optimization. Accuracy, balanced accuracy, macro F1, log loss and confusion matrices are recorded. A 2,000-resample test-row bootstrap, seed 42, provides a descriptive 95% interval for accuracy.

Coefficient arrays are saved as NPZ files with separate JSON schemas containing column order, category order, class order and selected C. This avoids serializing executable Python objects. Their intended purpose is reproducibility and inspection, not deployment as a clinical assessment service. The website does not apply these classifiers to its simulation controls.

10 ML results and interpretation

Measure	West to North	North to West
Training records	131	86
Test records	86	131
Selected C	1	10
Accuracy	97.67%	84.73%
Balanced accuracy	97.44%	86.69%
Macro F1	0.97714	0.84581
Log loss	0.11522	0.35780
Bootstrap accuracy interval	94.19-100.00%	78.63-90.84%
Majority accuracy	39.53%	45.04%
Fixed-C accuracy	97.67%	88.55%
Fixed-C log loss	0.20411	0.37376

Confusion matrices use true classes as rows and predicted classes as columns, in the order Vata, Pitta, Kapha. West to North gives rows [34,0,0], [2,24,0], [0,0,26]. North to West gives [45,14,0], [1,29,1], [0,4,37]. The asymmetry is material: transferring from the smaller Northern cohort produces more classification errors, especially Vata cases assigned to Pitta.

The log-loss-selected model improves held-out log loss relative to the fixed-C comparator in both directions. Accuracy is unchanged in West to North and lower in North to West. This tradeoff is consistent with optimizing probability quality rather than the number of top-label hits. It would be misleading to describe the revised classifier as uniformly more accurate or select a different model after seeing target results and still call those results untouched validation.

The bootstrap intervals condition on the trained model and resample test rows. They do not include training uncertainty, hyperparameter-selection variability, recruitment bias, site effects or label uncertainty. A high classification score on selected extreme types does not establish accuracy on mixed types, continuous composition, daily imbalance or treatment response. The underlying phenotypes may closely reflect how the constitution labels were assigned; reproducing those labels is not an independent biological test.

The legitimate improvements are stricter preprocessing separation, explicit training-only model selection, probabilistic evaluation, baselines, saved model coefficients and transparent reporting of unfavorable comparisons. The ODE's mathematical improvements should be assessed using stability and numerical tests, not by borrowing these constitution-classification scores.

Cross-cohort classification errors

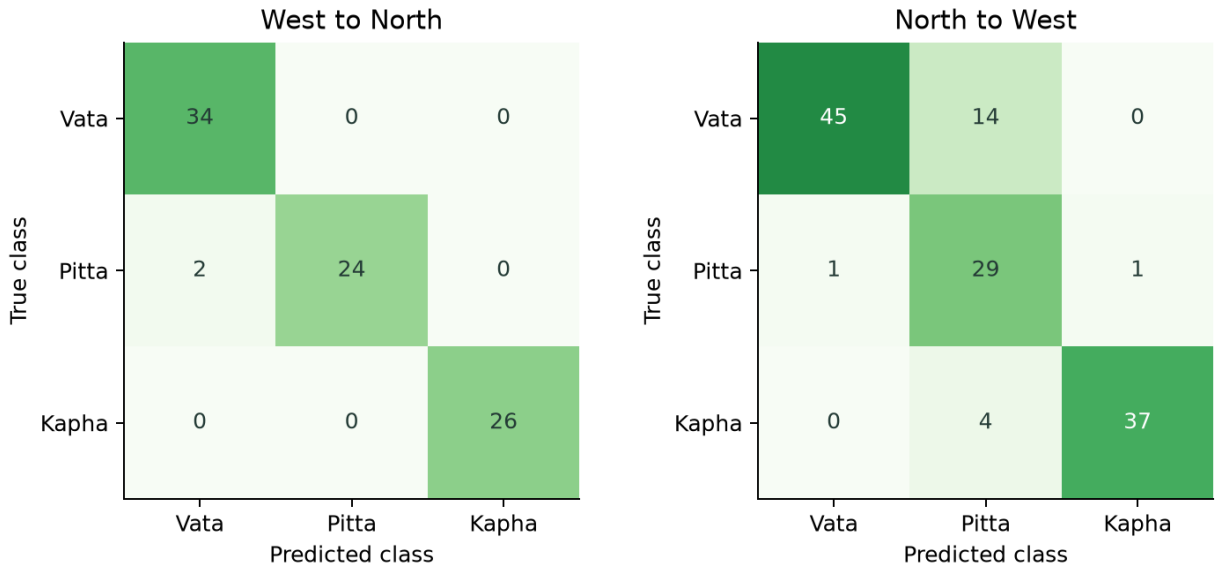


Figure 2. True classes are rows; predicted classes are columns. Each cell is a record count from the untouched target cohort. The pronounced Vata-to-Pitta error in North-to-West transfer illustrates the asymmetry between cohorts.

These confusion matrices belong to the constitution benchmark. They do not measure the accuracy of the dynamical system or the effect of negative feedback.

11 Website design and interaction

The interface uses a light cream background, white panels, restrained green accents and separate blue, terracotta and olive encodings for the three doshas. A serif page heading gives the laboratory an editorial identity, while system sans-serif fonts keep inputs, metrics and labels clear. Assets are local, and there is no dependency on external font delivery or stock photography.

The desktop layout places experiment setup beside the results. A persistent navigation rail links to simulation, mathematical explanation and validation evidence. The result hierarchy is deliberate: stability and recovery summaries appear first, the full trajectory follows, and composition geometry and time inspection provide detail. A short research notice remains visible near the top without interrupting each interaction.

The user selects a scenario, edits controls and chooses Run simulation. The browser sends a validated request to Python; it does not run a competing JavaScript approximation of the mathematics. A busy state disables the run button, successful completion updates all charts and summaries together, and errors are announced through a status region. If settings change afterward, the status explains that the results require rerunning.

The time slider updates the selected state and triangle marker without rerunning the ODE. Export includes the settings of the last completed experiment, even if the form has subsequently changed. Reset restores the default settings and reruns the experiment. A pulse that ends beyond the selected horizon is rejected before submission, with matching server-side validation for direct API callers.

Responsive CSS moves the navigation above the content on smaller screens and stacks setup and outputs on mobile. The composition diagram remains available at intermediate widths. Native labelled controls, visible keyboard focus, semantic sections, descriptive SVG labels and an aria-live status region provide a basic accessibility foundation. This is not a formal WCAG certification; screen-reader and contrast audits across all devices were not completed.

Browser verification covered actual Python-backed rendering, the equilibrium scenario, state scrubbing, pulse-window rejection, reset, export and responsive width. The preview image included in the deliverable is a captured rendering of the implemented website, not a design mockup. All shown scientific data originate in the local API or saved experiment outputs.

12 Verification and acceptance evidence

The active suite contains 25 deterministic pytest cases, including parameterized cases. It checks exact equilibrium preservation; positive normalized shares and centered deviations; agreement with the analytic matrix-exponential limit; negative Lyapunov derivative over 200 seeded states; faster default recovery with feedback; all three perturbation directions; unchanged baseline; the Aitchison distance identity; identical trajectories at zero gain; and null recovery when the horizon is insufficient.

Input tests reject zero or non-normalized baseline shares, nonfinite initial values, invalid pulse windows, negative forcing and invalid decay. Numerical tests compare final states across two reporting grids and ensure pulse discontinuities appear in the output grid. Additional tests check the bounded dissipative projected controller, the two-dimensional tangent basis, invariance to common log offsets and finite output at the API's strongest allowed rates.

API tests verify the static entry page, health endpoint, successful simulation and HTTP 422 responses for invalid or unknown inputs. Tests run locally through FastAPI's TestClient without external data services. The suite emitted one dependency deprecation warning about the test client's HTTP transport; it did not fail any test or affect the runtime simulation. Exact test output and JUnit XML are included in results/.

Eleven retained baseline tests passed separately. The baseline weather API smoke and forecast tests were excluded from that offline run. The new web server does not call those modules, so they are not hidden dependencies of the active application. The complete legacy_v1 suite, public hosting, authentication, load tests, security penetration tests and cross-browser certification were outside this acceptance scope.

Browser interaction testing checks the integrated path that unit tests cannot cover. The equilibrium scenario returned zero deviation and the baseline proportions. Moving the scrubber to day 10 updated the displayed composition and distance. A 30-day pulse within a 21-day horizon produced a clear validation message. Desktop and mobile viewport checks were performed, including a DOM overflow check at 390 pixels. The JavaScript file also passed Node's syntax check.

These tests provide evidence of software and mathematical correctness under the exercised cases. They cannot certify the traditional framework as a biological theory, prove the expert assumptions, or substitute for clinical study. Numerical accuracy, usability, empirical calibration and clinical utility remain distinct validation targets.

13 Architecture and API contract

The system has three layers. web/index.html, style.css and app.js implement presentation and interaction. lab/server.py validates requests and serves the website through FastAPI. lab/model.py contains the ODE, composition mapping, solver and performance calculations. The offline lab/experiments.py workflow creates numerical and ML evidence independently of browser interaction.

GET /api/health returns server and model status. POST /api/simulate accepts baseline, initial, days, decay, coupling, gain, amplitude, target, start and duration. It returns controlled and natural objects with time samples, centered deviations, composition shares, distances, centered control vectors, baseline, parameters, local stability and summary metrics. GET /api/evidence returns the saved experiments JSON or a clear 404 if experiments have not been generated. GET /docs exposes generated API documentation.

The request schema forbids unknown fields and nonfinite numbers. Baseline and initial arrays must have exactly three elements; decay has exactly three positive elements no greater than 2. Gain is limited to 0-3, coupling to 0-1, amplitude to 0-3 and horizon to 1-90 days. Model-level validation checks that the pulse lies inside the horizon and baseline shares sum to one. Errors are returned as HTTP 422 rather than producing invalid scientific output.

The application binds to 127.0.0.1 on port 8000. It is intended for a single local research user. It has no authentication, durable accounts, cross-user isolation, job queue, rate limiter, database or production deployment configuration. The active service stores no personal data. Result export is a client-side download; experiments overwrite only the derived local result files when deliberately rerun.

The new core does not import the original Engine. That boundary avoids inheriting live-weather calls and unrelated tracking behavior. The original data are read only by the offline cohort benchmark. The baseline source and attribution files are included to preserve provenance and enable future work, but are not presented as part of a newly validated production application.

14 Installation and reproducibility

The project was exercised on Windows with Python 3.14.2. Direct dependencies are pinned in requirements.txt, and the installed package versions are recorded in results/environment.txt. The package contains source and derived results rather than a Python interpreter or copied environment. First installation requires downloading packages; normal simulation and website use are then offline.

Open a terminal in the extracted project directory and create a virtual environment. On Windows run python -m venv .venv, then .venv\Scripts\python -m pip install -r requirements.txt. Start the application with .venv\Scripts\python -m lab.server. Open http://127.0.0.1:8000 in a browser and leave the terminal running. Ctrl+C stops the server. On macOS/Linux use .venv/bin/python instead; those operating systems were not verified in this delivery.

Run .venv\Scripts\python -m pytest tests -q --junitxml=results/tests.xml to reproduce the active test suite. Run .venv\Scripts\python -m lab.experiments to regenerate the solver comparison, gain sweep, parameter sensitivity, synthetic calibration, cohort benchmark and fitted coefficient/schema files. The precomputed results are included, so the website evidence panel works immediately after installation without retraining.

Key outputs are results/experiments.json, results/default_trajectory.json and results/default_trajectory.csv. The JSON preserves all default states and controls, while the CSV provides time and controlled/natural shares for external plotting. NPZ coefficient files and JSON schemas retain model class/category ordering. Seed 91 controls synthetic observation noise; seed 42 controls cross-validation and bootstrap sampling.

Do not launch web/index.html by double-clicking it: it expects same-origin Python API endpoints. If port 8000 is occupied, stop the other process or run uvicorn lab.server:app --host 127.0.0.1 --port 8001 and use the corresponding URL. If the evidence file is missing, rerun the experiment module. If package installation fails, verify Python version and network access before changing the pinned environment.

The source package includes the report in PDF and editable Markdown, numerical figures, the website preview and test logs. A SHA-256 checksum is supplied for the final ZIP. The archive excludes virtual environments, caches, personal runtime databases, temporary files and installed dependencies. It preserves original source/data attribution rather than assigning an unsupported new license to the input project.

15 Limitations and defensible conclusions

The ODE equations are a proposed mathematical formalization. Their parameters, coupling topology, pulse definition and feedback law were selected to support transparent analysis, not derived from a validated physiological mechanism. Baseline p is fixed within each experiment but may be manually specified; this does not establish that a person's equilibrium can be measured accurately through a questionnaire.

The stability proof is conditional on positive decay, nonnegative coupling and gain, the stated instantaneous controller and the model's smooth dynamics. Delays, asymmetric or destabilizing couplings, unknown disturbances, measurement errors and treatment constraints could change the problem. The model assumes the state is available to the controller and does not implement a state observer. Its smooth bounded controller is not a clinically actionable treatment policy.

Synthetic identification uses data generated by the same model family being fitted. That makes it a useful implementation test but an optimistic evaluation of empirical model adequacy. A full-rank local Jacobian does not guarantee global identifiability, useful confidence intervals or robustness to a misspecified observation process. Coupling is fixed specifically to avoid a known structural confounding.

Cohort classification has different limitations: selected extreme types, small sample sizes, possible site shifts, uncertain feature semantics and labels related to the phenotype definitions. Cross-cohort performance cannot validate trajectories, mixed compositions or intervention effects. The benchmark retains the less favorable accuracy result in one direction rather than hiding it behind an aggregate number.

The project can defensibly claim an implemented compositional ODE model; an explicit bounded negative-feedback mechanism; a stability proof under stated assumptions; numerical agreement with an exact linear limit; reproducible perturbation and recovery experiments; synthetic recovery of known parameters; a leakage-reduced cohort benchmark; and a tested local interactive website.

Further scientific work should validate measurement procedures, collect appropriate longitudinal observations, preregister empirical evaluation and compare against simpler models on untouched participants or sites. These are research requirements for stronger claims, not extra product features added to this delivery. No clinical diagnosis, treatment efficacy, optimal dosage, or calibrated human recovery forecast is claimed.

16 Sources and provenance

The two supplied inputs are dosha_v2_codebase.zip and Dosha Forecasting and Tracking App Data, ML and Modelling Plan.docx. The archive provides the baseline implementation and datasets; the plan provides scope history, draft mappings and proposed research directions. Its embedded suggestions are not treated as independent user instructions or as verified empirical facts.

The model construction and stability proof in this report are derived for the implemented equations. General background on Lyapunov analysis, state-space models and feedback is available in Astrom and Murray, Feedback Systems: An Introduction for Scientists and Engineers, Caltech repository: https://authors.library.caltech.edu/records/yzs24-xsx88 . This source supports the control-theory framework, not the dosha interpretation.

SciPy's solve_ivp documentation describes the adaptive initial-value solver and its tolerance/output controls: https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html . The least_squares documentation describes bound-constrained nonlinear least squares: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.least_squares.html . These technical references were consulted for the numerical implementation.

The original dataset attribution identifies Mendeley Data, Prakriti constitutions of North Indian and Western Indian cohorts, DOI https://doi.org/10.17632/dyx5cgxdrf.1 , and Tiwari P, Kutum R, Sethi T et al., PLoS ONE 12(10), e0185380 (2017). Dataset licensing and publication details in this package are retained from the supplied attribution; no new independent clinical validation was performed.

Original activity and herb attribution files remain alongside their baseline data. Those data are not inputs to the new ODE experiments. The report does not reproduce copyrighted questionnaire manuals or convert the inferred feature dictionary into an assessor instrument. Numerical tables are generated from this project's included result files, and the website image is captured from the implemented local application.